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RESEARCH ARTICLE

Creation and application of a geometric digital twin model for construction of spatial cable structures: A case study

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ABSTRACT This paper presents a dynamic geometric digital twin creation method for cable-net structures in construction which creates the building information model (BIM) for construction process visualization and finite element model (FEM) for cable shape and force prediction. The basic dynamic geometric information for parametric BIMs and FEMs is extracted from multi-stage laser scanning during construction. Multi-stage BIMs are presented as Industry Foundation Classes (IFC) format where the cables are created by approximating the curved shapes with straight segments. Complex components like cable clamps are imported as Revit Family and replicated at the measured locations. Multi-stage FEMs utilize the ABAQUS secondary development technology to parametrize geometric configuration, loads and constraints between components based on actual connections. A continuous dynamic BIM from one stage to another can be created based on construction progress and the deformed shapes derived from synchronously updated FEM calculations. Total duration for modeling and computation of each stage in construction is within 140 s. The average distances between the multi-stage BIMs and point clouds are within 15–20 mm. The prediction deviations of cable shapes and forces between FEM results and the measured data of the next stage are 1.2% and 1.5%, which considers temperature and construction progress.

KEYWORDS geometric digital twin, spatial cable structure, FEM, BIM, point cloud

1 Introduction

Pre-stressed cables are widely utilized in large-span spatial structures due to their lightweight, high strength, and flexibility [1]. During the construction process, cables, as the primary load-bearing components, often experience significant changes in cable shapes and forces. Therefore, monitoring cable forces and shapes during construction have a substantial impact on construction sequencing decisions and safety [2,3]. Traditional structural health monitoring typically involves placing contact sensors at key locations of the structure to collect relevant data [4]. Although the monitoring method can obtain state information from specific points, it often fails to provide insights into the overall mechanical response and geometric shape of the structure. Additionally, visualizing the deformed geometry of the structure after loading is similarly challenging.

As a comprehensive digital technology, digital twin (DT) technology offers excellent visualization capabilities and accurately reflects the physical state of entities, enabling various fields to accelerate their digital

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transformation [5,6]. In the civil engineering domain, the DT of a structure reflects the structural behavior in a timely manner based on the sensor data and simulation which improves the construction quality and structural safety throughout the life cycle of the structure [7]. The application of DT technology during the construction phase and subsequent operational maintenance stages can significantly improve the level of automation and decision management in the construction industry [8]. With advancements in Building Information Modeling (BIM), the Internet of Things (IoT), and numerical simulation techniques, research on the application of DT technology in the construction field has markedly increased, primarily focusing on the operational and maintenance phases [9–11]. In cable structures, Wang et al. [12] created a DT model through finite element model (FEM) which reflects the physical structure of a cable dome. A structure damage sample database is established using the DT model and the hierarchical deep learning method is implemented for the intelligent identification of damage in the cable dome structure. Moreover, they [4] created several approximate low-order models based on the scaled model to accelerate the calculation process and constructed a DT platform for visualization and interaction through Unreal Engine 4. Liu et al. [13] and Shi et al. [14] proposed the method of creating high-fidelity FEM as digital models. The model is updated by parameter optimization to realize the error minimization between FEM results and mechanical response of the structural entity. An intelligent safety evaluation method for cable-net structures [15] is also proposed by generating a large data set calculated by a high-fidelity FEM with key factors adjustment and a support vector regression model. Li et al. [16] proposed a Prognostics and Health Management (PHM) system for FAST cable-net structure based on DT technology. The high precision sensors and measuring equipment promote to construct the high-fidelity model. The structural geometry and environment data are updated in the DT model periodically and the fatigue life of components is predicted. However, the number and types of components will change during the construction process of spatial cable structures. It is unrealistic to mimic the operation and maintenance phase by establishing only one geometric model for calculations. Additionally, the deformations of the cable structures need to be continuously updated. Traditional cable structures cannot record deformations at every point like the FAST cable net, where each point is equipped with actuators for monitoring displacement. Manually extracting data from dozens or hundreds of cable joints still remains a labor-intensive task.

In the process of simulating physical systems using virtual models, the need to solve partial differential equations (PDEs) leads to a significant increase in computational costs when dealing with complex operating conditions [17]. To meet the real-time feature of DT models, operator learning and physics-informed neural solvers are employed for real-time response prediction of the models [18]. In this approach, the objective function is supplemented with the minimization of the energy equations of PDEs for deep learning model training [19]. This not only retains computational accuracy but also effectively improves computational performance, while addressing the limitation of poor generalization in purely data-driven models [20]. However, in practical scenarios, establishing a high-fidelity virtual model serves as the foundation for ensuring simulation accuracy and training accuracy. Compared with the operation and maintenance phase, the types and quantities of model components as well as model boundaries during the construction phase often change with the construction progress. Therefore, it is an effective approach to realize DT technology to first ensure the efficiency and accuracy of multi-phase modeling, and then combine it with deep learning methods to improve analysis efficiency.

In the process of creating a DT model, the acquisition of a geometric model that accurately reflects the real physical world often serves as the foundation for the virtual twin model. Three-dimensional laser scanning technology, with its high precision, speed, extensive measurement range, and high scanning resolution, produces point cloud models that are generally regarded as virtual representations of real physical geometries [21]. However, the raw point cloud data only carries geometric information for each point (such as coordinate data and normal vector) and color information. The information is not readily usable by engineers or administrators and cannot be integrated with other information in a DT model [22]. Over the past few years, researchers have focused on the automatic conversion of point cloud data into geometric representations, BIM models, or FEM geometries [23]. This process is known as the creation of geometric digital twin (gDT) models [24].

In the field of buildings, Drobnyi et al. [25] proposed a method of digitizing the geometry of Manhattan-world buildings by segmenting the entire indoor space through detecting empty regions and expanding the regions. The volumetric representations of walls, windows and doors detected from indoor space are reconstructed. Mahmoud et al. [26] proposed an automated BIM generation workflow for large-scale indoor environment through deep learning semantic segmentation and BIM parametric modeling. Xue et al. [27] transformed the BIM reconstruction into a multimodal optimization problem by registering the point cloud with the parametric BIM family. Considering that building components are typically made up of slender members, the extraction of component axes can serve as the basis for structural geometric representation [28] or BIM reconstruction [29].

In the field of bridge and transportation, Pan et al. [30] segmented the pavements into a more detailed level (lanes, hard shoulders and central reserves) on the basis of semantic segmentation of road surfaces. A graph is generated based on the topology of highways to represent the highway components and their relationships. Lu and Brilakis [31] proposed an automatic geometric DT creation method for reinforced concrete bridges by slicing-based Industry Foundation Classes (IFC) object fitting, which improved modeling accuracy and efficiency. Hu et al. [32] proposed several fitting methods to create the geometry of arch ring and other arch bridge components. A translation strategy is implemented to solve the issue of lacking cross-sectional parts. The gDT is generated and visualized in CATIA. Mafipour et al. [33] created the parametric prototype models (PPMs) to represent the bridges, and the geometric model is generated automatically by optimizing the parameters of PPMs to fit the point cloud through metaheuristic optimization. Shu et al. [34] proposed the BCR-Net to realize the component segmentation from bridge point cloud and the geometric model is created by the component geometric feature extraction and parametric modeling.

In the field of cable structures, Zhang et al. [35] proposed multimember central shrinkage algorithm and "3-point normal" cylinder fitting method [36] to extract the cable dome components from point cloud. The joint coordinate is detected through the intersection of component axes and the node adjacency matrix is created for topology connection relationship representation and deviation detection compared with a design model.

In the field of industry, CLOI-NET [37] is proposed for the class segmentation of industrial point cloud and CLOI-Instance graph connectivity algorithm [38] solves the instance segmentation problem of the industrial point cloud after classification which decrease the cost and manual labor for automatically generating geometric DT of industrial facilities.

Although there has been progress in automated geometric extraction and modeling based on point clouds, several limitations still need to be addressed.

1) Post-loading deformed point cloud geometry is unsuitable for simulation without internal force data.

2) Static point cloud geometry fails to support BIM/FEM updates as physical conditions evolve.

3) BIM and simulation models lack correspondence for synchronized updates.

Building on the fact that Limitation 1 has already been addressed in our previous research [39], this paper proposed an automated creation method of geometric DT model of cable structures in the construction process to solve Limitation 2&3 which is illustrated in Fig. 1. The method achieves two unique contributions: 1) the parametric synchronization of the geometry model in IFC and FEM based on the cable geometry; 2) the dynamic calculation and model update considering the physical environment.

This workflow realizes IFC parametric modeling and FEM parametric modeling based on common geometric data extracted from the point cloud (such as the cable shape, coordinates of cable anchors and cable clamps) and the construction information (such as connection relationship). Moreover, this study performed multi-stage scanning of the structure during construction and extracted geometric data for modeling. Considering that the point cloud model represents only a snapshot of the structural state, this method conducts finite element calculations based on construction progress records and temperature monitoring data, synchronously updating the computed deformed geometric data in the IFC model to address the gaps in model states between adjacent scan stages.

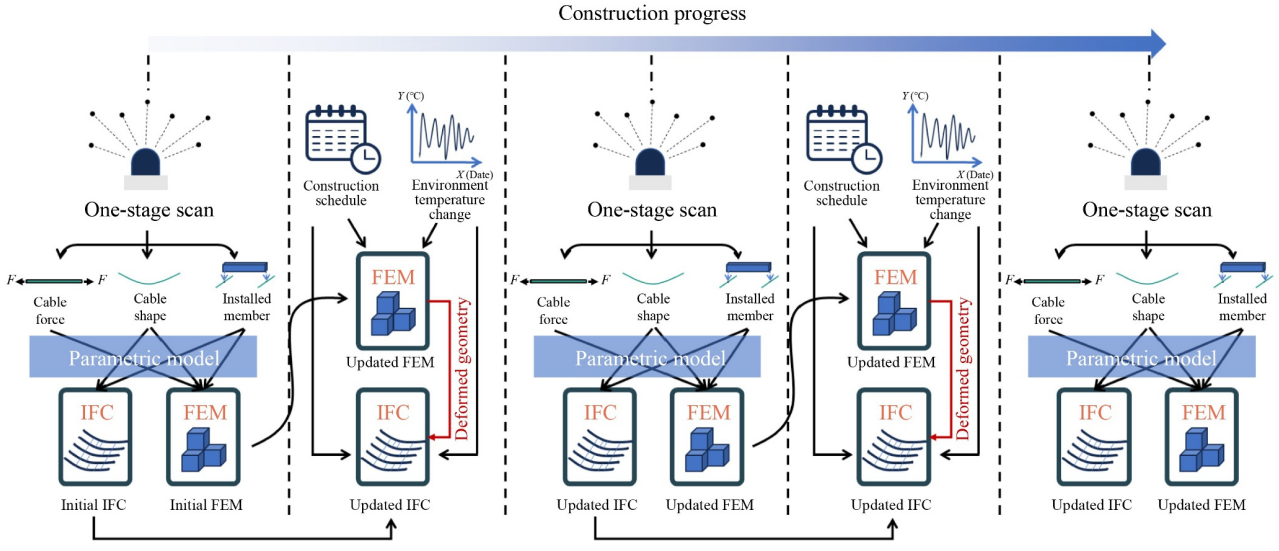


Fig. 1 The workflow of method.

2 Geometric digital twin model creation for cable structures

2.1 Industry foundation classes model creation of cable structures for one-stage scan

The information extracted from point cloud includes the point coordinates of cable shapes, cable clamps, cable anchors and cable forces [39]. Based on the information, the IFC model can be reconstructed for each construction stage of the structure by using the geometric relationships between cable clamps, connecting members, and anchor points.

IFC data, as a universal data format for BIM models, is an open, object-oriented data model used for describing, exchanging, and sharing information in the architecture, engineering, and construction (AEC) industry [40]. Modeling and updating based on the initial IFC model facilitate information transfer across different stages and software. IFCOpenShell, an open-source software library, can be used to process IFC data [41]. Given that structures involve numerous key coordinate points and similar components, parameterized modeling of IFC data enables efficient creation and updating of BIM model components.

2.1.1 Basic industry foundation classes creation and single cable modeling

Single cable is composed by the cable itself, cable anchor ends, ear plates and cable clamps. The cables have a circular cross-section which can be parametrically modeled using extrusion operations. However, the cable anchor ends, ear plates and cable clamps are complex geometric components that cannot be easily modeled using simple parameterization. Therefore, detailed reconstruction of the cable anchor and clamps was carried out in Rhino software based on CAD drawings as shown in Fig. 2. These detailed components are then imported into Revit software as new families.

During the import process of these new families, it is essential to define the origin of the new family components. In subsequent modeling, changes to these components involve rotations and translations around their origin points. For the ear plates and cable anchor ends, the origin is set at the center of the pin shaft. For cable clamps with two parallel cables passing through as shown in Fig. 2, the origin is set in the middle of two holes. For cable clamps with only one cable passing through, the origin is set at the center of one hole. The schematic diagram of the origin settings is shown in Fig. 2. At the initial stage, the created geometries for the ear plates, anchor points, and cable clamps are imported into the Revit model and exported as the IFC4 reference view [Architecture] data format. Subsequent reconstruction of the cable-net model can be performed using parameterized modeling based on this basic IFC.

Based on the basic IFC, the cable itself should be constructed first for the orientation confirmation of other components. Considering the curved shape of cables, modeling is performed by approximating the curved shape with straight segments. The entire cable shape is divided into multiple straight segments. For instance, considering a segment of the cable AB as shown in Fig. 3, a cylinder with a radius and a length of l_{AB} is created with its starting point at [0,0,0] and oriented along [0,0,1]. The unified rotate matrix [42] is defined by Eqs. (1) and (2). This paper defines the coordinate system for the cable structure, with the y-axis horizontally along the direction of the cables and the z-axis perpendicular to the ground, as shown in Fig. 3. $\overrightarrow{p_1}$ is the origin vector of the initial component which is [0,0,1] in the cable modeling. $\overrightarrow{p_2}$ is the target vector which is $\overrightarrow{AB}$. θ is the angle between $\overrightarrow{p_1}$ and $\overrightarrow{p_2}$. Translate $\boldsymbol{T}$ is from the origin (0,0,0) to point A which is equal to the coordinate of A. Finally, the affine transformation $\boldsymbol{M}$ is assembled by Eq. (3) and the cable segment is transformed by "geometry.edit_object_placement" in IFCopenshell. All cable segments can be unified parameterized into a whole cable. For other components, all the parametric transformations can be similarly

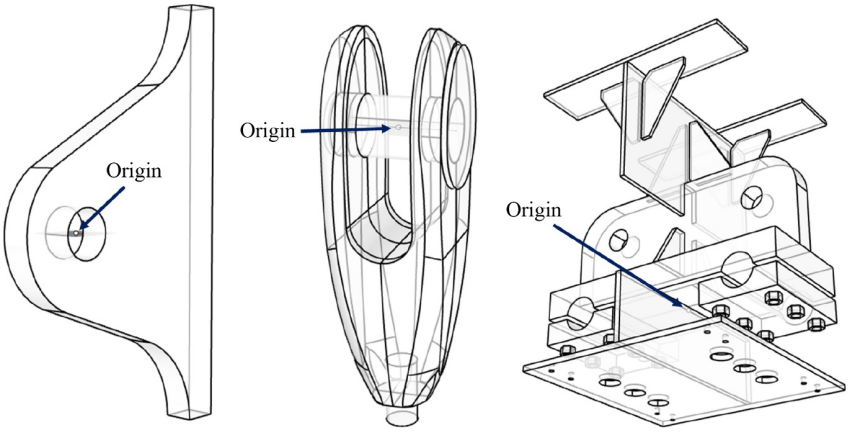


Fig. 2 Complex components and setting of the origin.

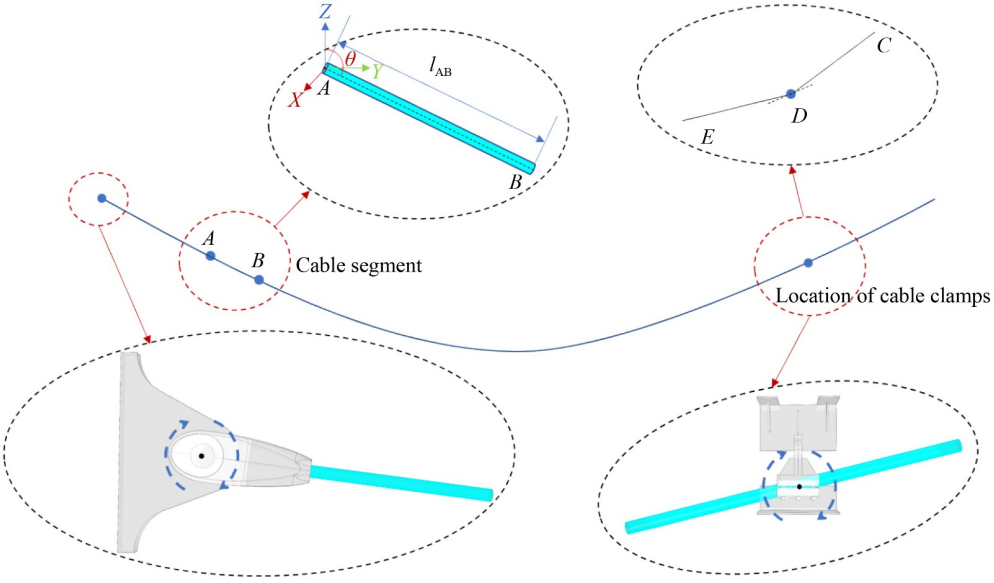


Fig. 3 Cable diagram.

derived by extracting the transformation variables $\overrightarrow{p_1}$, $\overrightarrow{p_2}$, and $\boldsymbol{T}$ based on the geometric relationship between the components and the cables.

$$\boldsymbol{n} = \frac{\overrightarrow{p_1} \times \overrightarrow{p_2}}{\left\| \overrightarrow{p_1} \times \overrightarrow{p_2} \right\|}, \tag{1}$$

$$\boldsymbol{R} = \begin{bmatrix} \boldsymbol{n}_x^2(1-\cos\theta)+\cos\theta & \boldsymbol{n}_y\boldsymbol{n}_x(1-\cos\theta)-\boldsymbol{n}_z\sin\theta & \boldsymbol{n}_z\boldsymbol{n}_x(1-\cos\theta)+\boldsymbol{n}_y\sin\theta \\ \boldsymbol{n}_x\boldsymbol{n}_y(1-\cos\theta)+\boldsymbol{n}_z\sin\theta & \boldsymbol{n}_y^2(1-\cos\theta)+\cos\theta & \boldsymbol{n}_z\boldsymbol{n}_y(1-\cos\theta)-\boldsymbol{n}_x\sin\theta \\ \boldsymbol{n}_x\boldsymbol{n}_z(1-\cos\theta)-\boldsymbol{n}_y\sin\theta & \boldsymbol{n}_y\boldsymbol{n}_z(1-\cos\theta)+\boldsymbol{n}_x\sin\theta & \boldsymbol{n}_z^2(1-\cos\theta)+\cos\theta \end{bmatrix}, \tag{2}$$

$$\boldsymbol{M} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{T} \\ 0 & 1 \end{bmatrix}. \tag{3}$$

2.1.2 Construction of complex components

As mentioned in the previous section, the basic IFC stores the geometry information of complex components. Therefore, the parametric process is to copy the complex component IFC information and arrange every component into the corresponding location.

For cable clamps, extract IFC information of cable clamp from the basic IFC and copy all the attribute information as shown in Fig. 4. Only the Globally Unique Identifier (GUID) of the new component should be recreated. The ObjectPlacement attribute will be modified in the following.

The clamp arrangement is not only to translate the component into the measurement position, but also to make the cable pass entirely through the clamp's predefined hole. The model of the clamp is designed with a hole aligned along the [0,1,0] as shown in Fig. 3, whereas the cable direction at different positions of the clamp may not be parallel to [0,1,0]. Therefore, the tilt angle of the clamp is required to define for different positions. Due to the self-weight of the clamp, the slope of the cables at both ends will change. During the modeling process, the tilt angle of the clamp is defined as the average of these slopes, represented in the diagram as $\frac{\overrightarrow{ED}+\overrightarrow{DC}}{2}$, which is the slope of the dashed line in Fig. 3. So $\overrightarrow{p_1}$ is the origin vector of initial component which is [0,1,0]. $\overrightarrow{p_2}$ is the target vector which is $\frac{\overrightarrow{ED}+\overrightarrow{DC}}{2}$. Translate $\boldsymbol{T}$ is from the origin (0,0,0) to point D which is equal to the coordinate of D in Fig. 3. Therefore, all clamp arrangement can be parametrized by Eqs. (1)–(3).

Since the anchor points only exist at the two ends of the

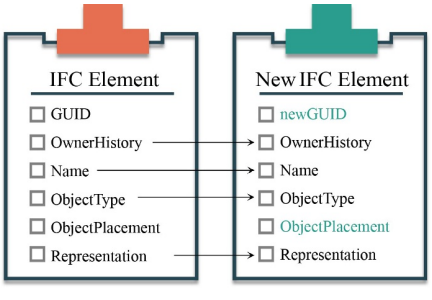


Fig. 4 IFC Element copy.

cable, each cable requires only two anchor points. First, copy two anchor ends and rotate their direction from [0,0,−1] ($\overrightarrow{p_1}$) to align parallel to the slopes of the cable ($\overrightarrow{p_2}$) at both ends as shown in Fig. 3. Translate $\boldsymbol{T}$ of the anchor is from the origin to the respective ends of the cables. The transformation of ear plates at both ends of the cable is also the similar to the cable anchor.

2.1.3 Construction of connecting members

In some cable structures, the connecting members are installed between adjacent cables during the construction phase, resulting in varying numbers of connecting members on different cables. It is necessary to automatically determine the number of connecting members on each cable. A quantity matrix is created to indicate the installation status of the connecting members between the cables at each stage.

The point cloud of cables is first projected into x–y space and segmented based on line extraction through Hough transform. Point clouds within a range l along a cable are segmented into the cable. For example, in Fig. 5, the length in the direction perpendicular to the cable will vary at different cable clamps due to differences in the number of installed connecting members. When connecting members are installed at both ends of a cable clamp, the length in the x-direction is approximately l. When connecting members are installed at only one end, the length in the x-direction is approximately $\frac{l}{2}$. Using these criteria, the number of connecting members to be installed at each cable clamp position can be quickly determined. The connecting member quantity matrix at every position of each cable clamp for the cable structure in Fig. 5 is shown in Table 1.

Considering connecting members usually have constant cross section, extrusion operation is also utilized for connecting member modeling. Parametric modeling of connecting members is realized based on the quantity matrices and design parameters at different stages. The installation status of the connecting members at the corresponding cable clamp between adjacent cables has been determined based on the quantity matrix. The installed connection member is parametrically modeled by querying the design parameter at the corresponding cable clamp position.

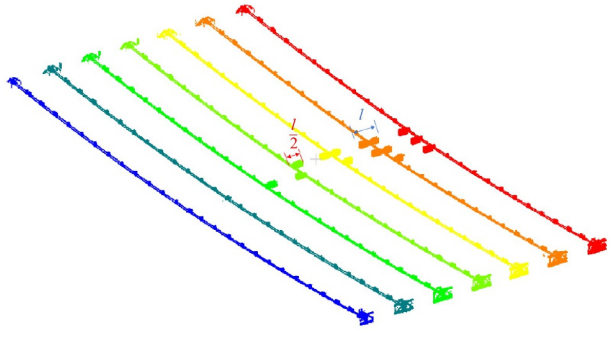


Fig. 5 The segmentation of cables and installation status decision.

Table 1 The connecting member quantity matrix

Clamp number	Cable 6	Cable 7	Cable 8	Cable 9	Cable 10	Cable 11	Cable 12
1	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0
12	0	0	0	1	2	2	1
13	0	0	1	1	1	2	1
14	0	0	0	0	0	1	1
15	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0
19	0	0	0	0	0	0	0
20	0	0	0	0	0	0	0
21	0	0	0	0	0	0	0
22	0	0	0	0	0	0	0
23	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0
25	0	0	0	0	0	0	0

During the modeling phase, use the connecting member quantity matrix to determine the positions for the connecting member modeling. For example, to model the connecting member at the mid-span (13#) between cable 8 and 9, start by creating a cross section with a direction vector of [0,0,1] which is $\overrightarrow{p_1}$ and a start point at [0,0,0]. The length equal to the distance from the 13# cable clamp origin position of cable 8 to the 13# cable clamp origin position of cable 9. Next, the objective vector $\overrightarrow{p_2}$ is from the 13# cable clamp position of cable 8 to the 13# cable clamp position of cable 9. Finally, translate $\boldsymbol{T}$ is from the origin to the coordinate position of the 13# cable clamp of cable 8. This method can be applied to model other connecting member positions similarly.

2.1.4 Model visualization

After the above process, the final IFC files are generated at different stages and the IFC visualization is shown in Fig. 6. The visualization platform is "Open IFC Viewer" developed by Open Design Alliance (ODA). It can be observed from the figure that the cables successfully pass through the predefined holes of the cable clamps, with the anchor end positioned tangentially to the cable end. Thus, the parametric modeling of the IFC model for one-stage scan is complete.

2.2 Reconstruction of parametric finite element models for one-stage scan

2.2.1 Overview and simplifications

From the geometric data, a FEM of the existing cable-net state can be reconstructed. This involves using the existing cable shapes for modeling, applying corresponding external loads such as cable self-weight and connecting member self-weight, and applying internal cable forces to achieve equilibrium in the structure. Due to the extensive geometric points and load applications involved in finite element modeling of cable-net structures, using a secondary development approach with Python in ABAQUS can significantly improve modeling efficiency. The finite element modeling process is similar to IFC model reconstruction, but some simplifications and differences are as follows.

1) The anchor ends are simplified to a constrained hinge point, and the weight of cable clamps are simplified to concentrated loads.

2) Unlike IFC models, FEMs need to consider constraints at the endpoints and between components. This aspect will be detailed further in the subsequent modeling process.

2.2.2 Parametric modeling and assembly

Some unified basic information could be defined first in code to serve as the foundational information for subsequent modeling such as part geometry and material properties. Only two kinds of parts are considered in FEM modeling: cables and connecting members while the cable clamps is considered as concentrate loads and the cable anchorage is considered as boundary constraints.

The cables are modeled similarly to the IFC model, using straight lines to represent curves. Both cables and members can be modeled using beam elements. For the cables, the material parameters are as follows: the equivalent cross-section is a circular section with the materials of density, Young's modulus and the thermal expansion coefficient. For the connecting members, the material parameters include the cross-section, density and Young's modulus. Due to the varying installation of the connecting members at different locations, the self-weight of the associated components may also differ.

The identification and installation of the connecting members are similar to the IFC assembly process. In ABAQUS, "mdb.models.rootAssembly.translate()" and "mdb.models.rootAssembly.rotate()" functions are provided for transformation of every part. Based on the transformation information during IFC modeling, the final assembly result is shown in Fig. 7. During the assembly process, each instance in assembly model needs to be named for subsequent parameterization, constraint application, and load application, to facilitate information query and localization as shown in Fig. 7. For example, every instance of cable is named as part-cable-{i}, i is the cable number in FEM. Every instance of member is named as part-member-{i}-{j}, i is the member on the index of cable clamp and j represents the index of the cable to which the member is connected at the starting end. Continuous naming numbers help facilitate the uniform application of functions through loops in subsequent processes.

2.2.3 Parametric constraints and loads

Since the cable anchor connections and cable clamp connections are both pin connections, rotational constraints in one direction are released during constraint

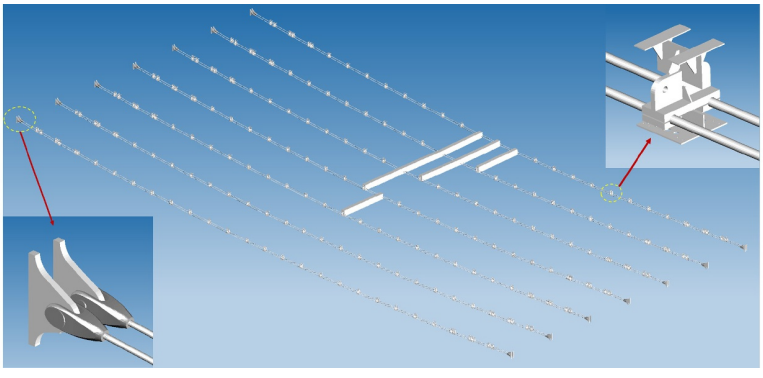


Fig. 6 IFC model on November 2, 2022.

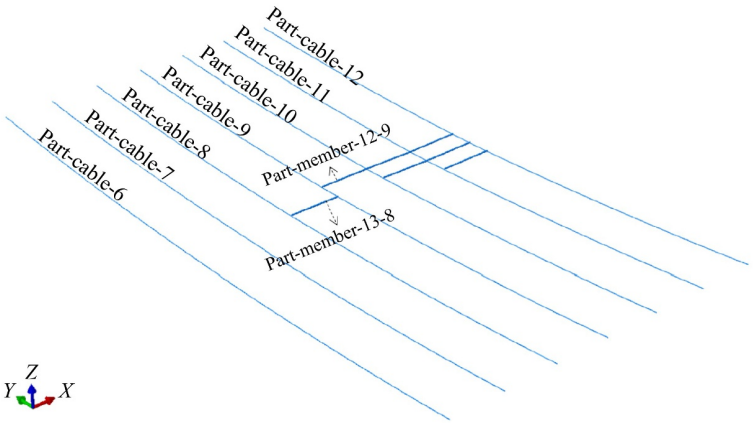


Fig. 7 Assembly of FEM.

definition. For the case in Section 3, the coordinate system for the cable structure is defined as follows: the *y*-axis along the direction of the cables, the *x*-axis along the direction of the connecting members, and the *z*-axis perpendicular to the ground. For anchor connections in ABAQUS, the rotation constraint is released about the *x*-axis and fixed about the *y*- and *z*-axes. For the example in Section 3, regarding the connections between connecting members and cable clamps, this type of rotation constraint involves interactions between two components. Specifically, coupling constraints are applied at the interface of each connecting member and cable clamp (defined via the Intersection Module), with rotation released about the *y*-axis and fixed about the *x*- and *z*-axes. The geometry of a cable is composed of multiple sequences of vertices. When applying boundary conditions at both ends, they can be directly applied to the first and last vertices. When applying the intersection between the connecting member and the cable, the corresponding two components can be located through their names and the connecting member quantity matrix. Then, constraints can be applied to the vertex sequence in the cable and the end vertex of the connecting member.

During the load application phase, self-weight loads are applied first in the entire domain. The concentrated loads from the cable clamps on internal vertices of every cable geometry. Since the geometry obtained from laser scanning represents the balanced state after the external loads and internal cable forces. To achieve the equilibrium state of the FEM, internal cable forces need to be applied to each cable. Cable forces are applied using the cooling method with a temperature reduction of

$$T = \frac{F}{Area \times E \times \alpha}, \tag{4}$$

where F is cable force, $Area$ is the effective cross-sectional area, E is Young's modulus of cables, α is the thermal expansion coefficient of cables.

2.3 Finite element model and industry foundation classes model update

As mentioned before, the geometry obtained from point clouds is a snapshot at a specific moment. As time evolves, the established BIM model geometry or FEM model geometry should be updated based on changes in physical conditions such as temperature and construction progress. At first, the FEM model based on one-stage scanning can be applied the temperature change load or installed connecting member self-gravity load on every cables. The temperature load can be applied by modifying the cable temperature. The newly installed connecting member can be updated by modifying the connecting member quantity matrix. The stress and deformation results of every cable can be solved after FEM updated and saved in ODB files. The deformation information is queried by instance name and added to the origin cable shape. The deformed cable shape and updated connecting member quantity matrix are used for updating the cable shape and geometric relationship with other components in IFC. The updated IFC model and FEM model reflect the new state of the structure in the physical environment. The updated process is illustrated in Fig. 8.

3 Project case

3.1 Project overview

The case study in this paper involves a single-curvature cable-net structure, which consists mainly of 18 sets of cables and connecting members between the cables. Each set of cables is composed of two parallel cables. The cables are connected to the connecting members through cable clamps, and the connections between the cables and the main structure are made via pins. The cable-net region and two types of connections are illustrated in Fig. 9. All the circular pipe connecting members are covered by curtain walls to form a rectangular prism. This paper

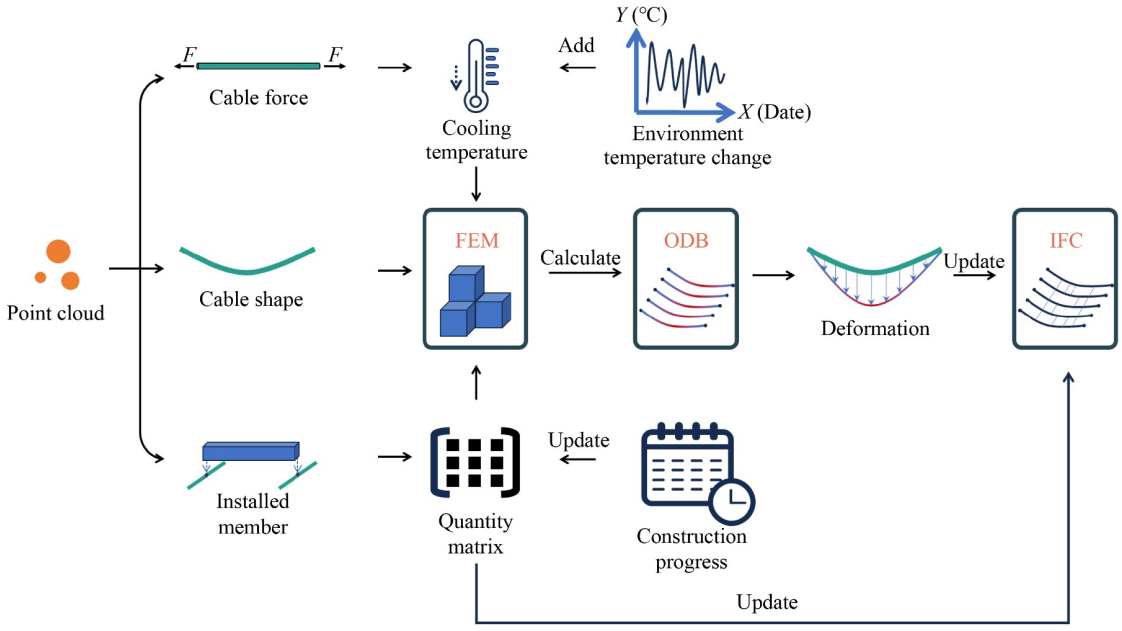


Fig. 8 FEM calculation and IFC update.

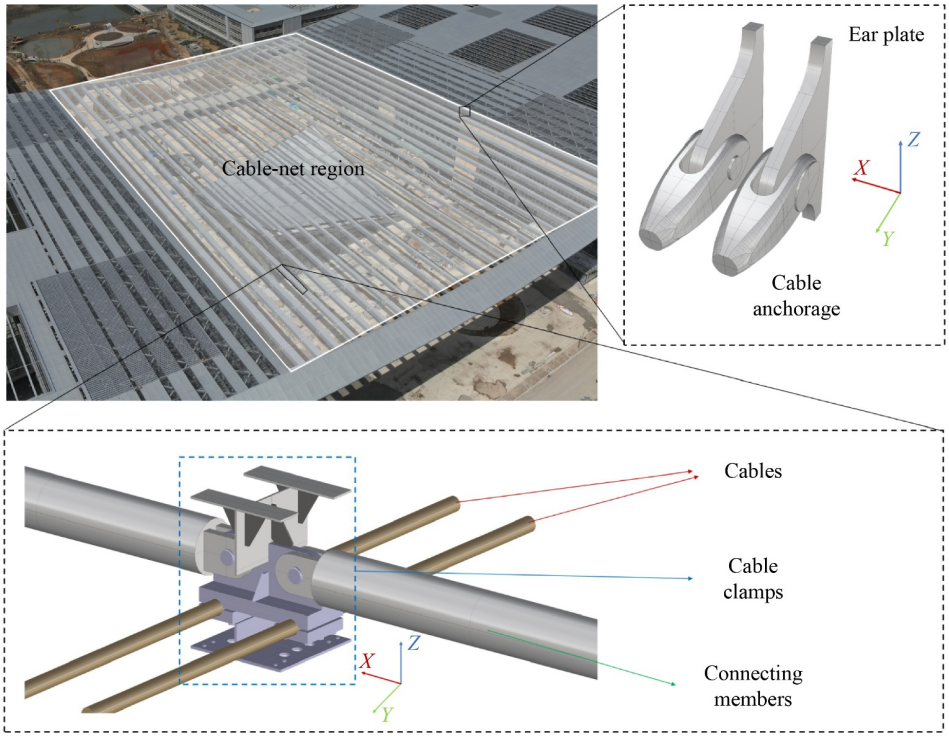


Fig. 9 Cable structure and connection method.

defines the coordinate system for the cable structure, with the *y*-axis along the direction of the cables, the *x*-axis along the direction of the connecting members, and the *z*-axis perpendicular to the ground, as shown in Fig. 9.

For the modeling process, the equipment and computational hardware are as follows: CPU Intel(R) Xeon(R) Gold 6254 CPU @ 3.10GHz, RAM 192GB, Operating System Windows 10, ABAQUS 2022 (Not use multiple processors), Python 3.9, Cloudcompare v2.11.3, IFCOpenShell v 0.7.0, open3d v0.16.0, numpy v1.26.4, Open IFC Viewer v25.5.0.

3.2 Construction process

The construction process for the cables is as follows: 1) install the cable clamps at the fixed positions of each cable set; 2) tension the cables; 3) install the connecting members after the adjacent cable sets have been

tensioned. The progress of cable tensioning and connecting member installation is shown in Fig. 10. During construction, two main control loads affecting cable shape are the self-weight of the connecting members and temperature loads. Temperature data, recorded by vibrating wire strain sensors installed on the steel structure, is shown in Fig. 11. The figure illustrates temperature fluctuations over time from October 30, 2022, to December 25, 2022, with noticeable daily variations and a sharp temperature drop around November 30.

3.3 3-Dimension (3-D) laser scanning and processing

During the construction, we conducted six 3-D laser scans of the structure on November 2, November 4, November 30, December 2, December 5 at the stage of construction and one scan of the as-built stage. The scanner equipment was the Leica P40. After scanning, the data was imported into Cyclone software for point cloud registration and noise removal. The results of these multiple scans are shown in Fig. 12.

According to the method proposed in previous work [39], the coordinates of all anchor points and cable clamps from the scanned point clouds at each stage are extracted. Subsequently, the cable shapes and cable forces are determined using the segmented catenary theory. The processing flow of every scan can be found in Fig. 13. The results for the mid-span sags and cable forces at each stage are presented in Tables A1–A6 at Appendix A in Electronic Supplementary Material.

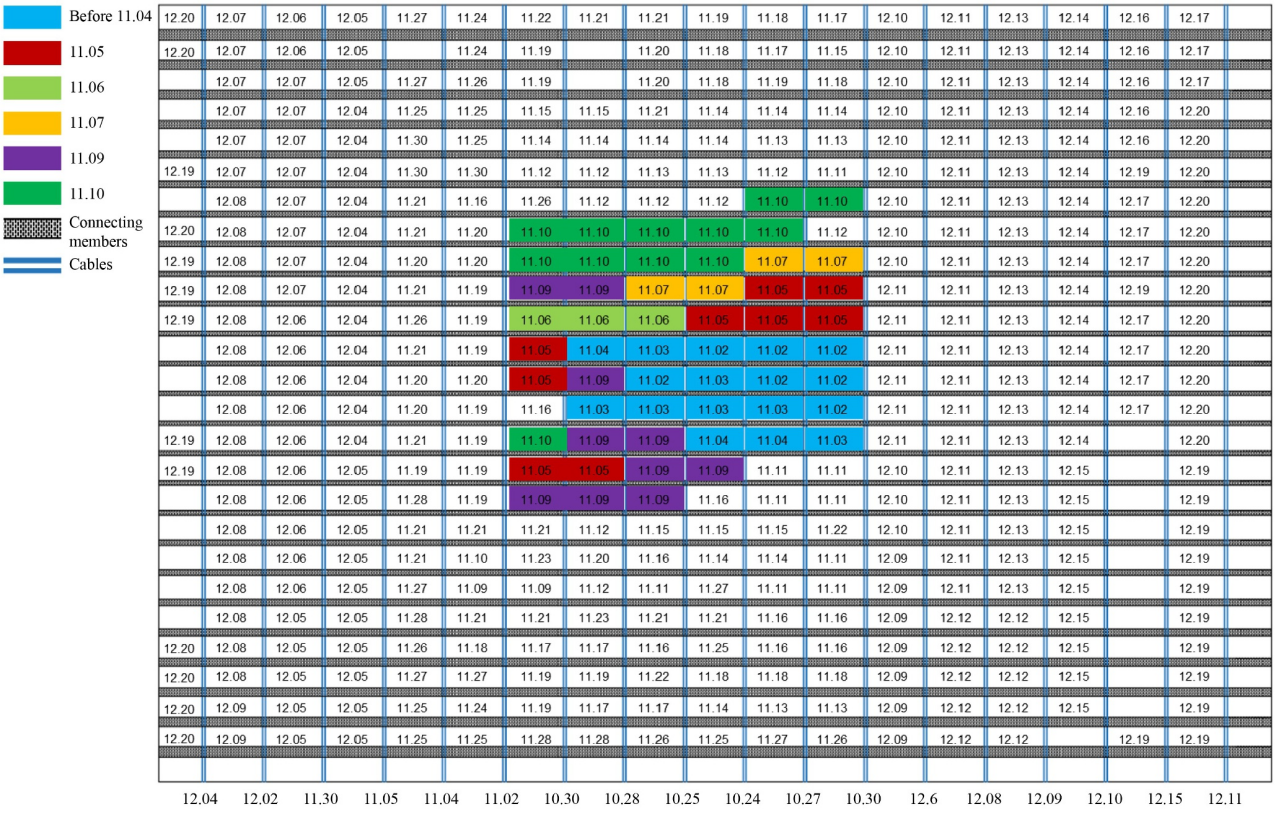


Fig. 10 Construction process.

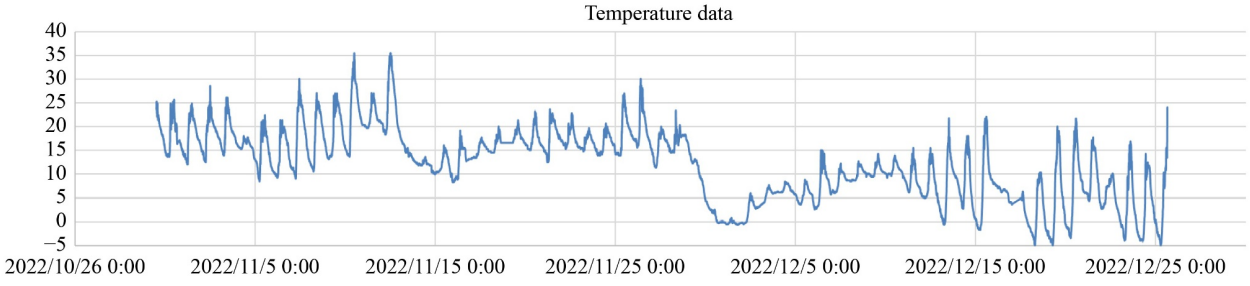


Fig. 11 Temperature curve.

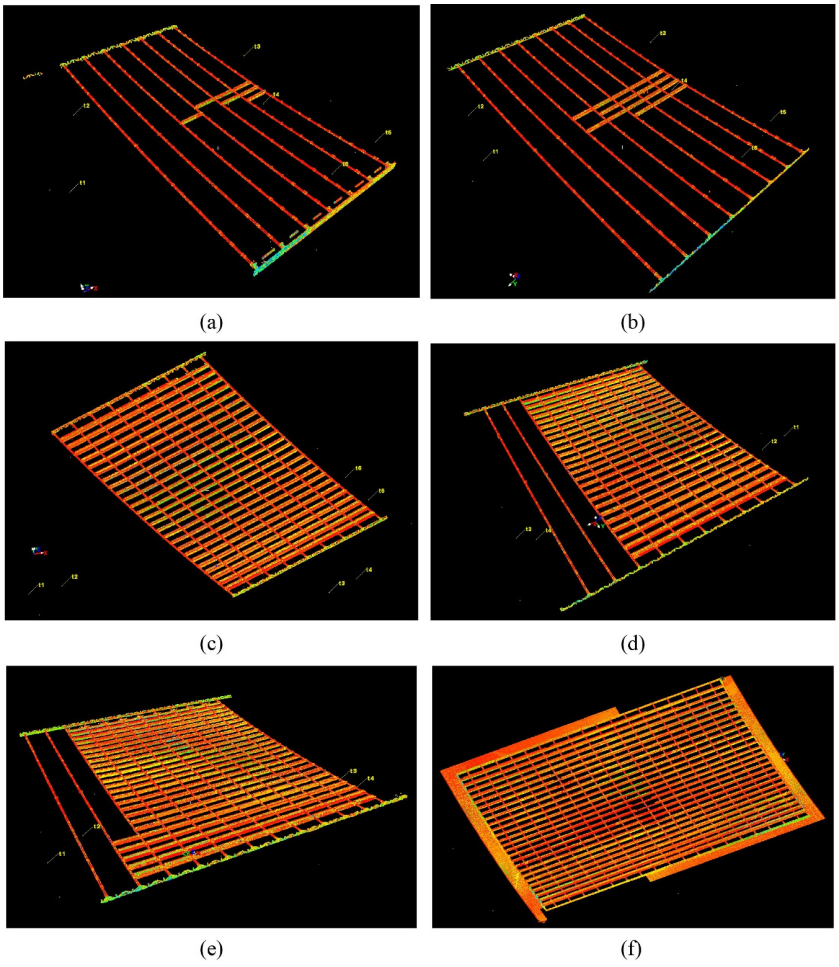


Fig. 12 Scan results of multi-stage: (a) scan results on November 2, 2022; (b) scan results on November 4, 2022; (c) scan results on November 30, 2022; (d) scan results on December 2, 2022; (e) scan results on December 5, 2022; (f) scan results at as-built stage.

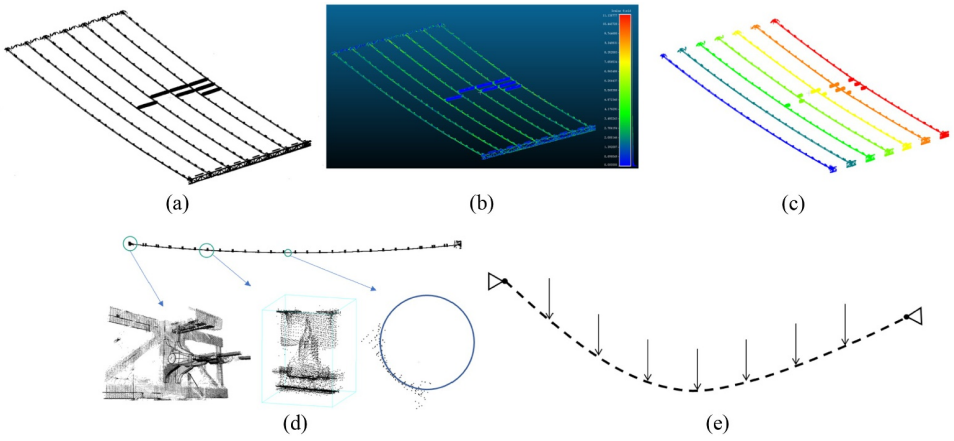


Fig. 13 Point cloud processing flow: (a) raw point cloud; (b) curvature filtering; (c) hough transform segmentation; (d) coordinate extraction; (e) segmented catenary theory.

4 Model verification and application

4.1 Industry foundation classes and finite element model from multi-stage scans

Based on the obtained cable shapes and cable forces, as well as the parametric IFC modeling approach, IFC model data for multiple stages can be generated. The model representations are shown in Fig. 14. Since finite element analysis focuses on the structural state while IFC models are concerned with visualization. The assumptions for finite element modeling are as follows.

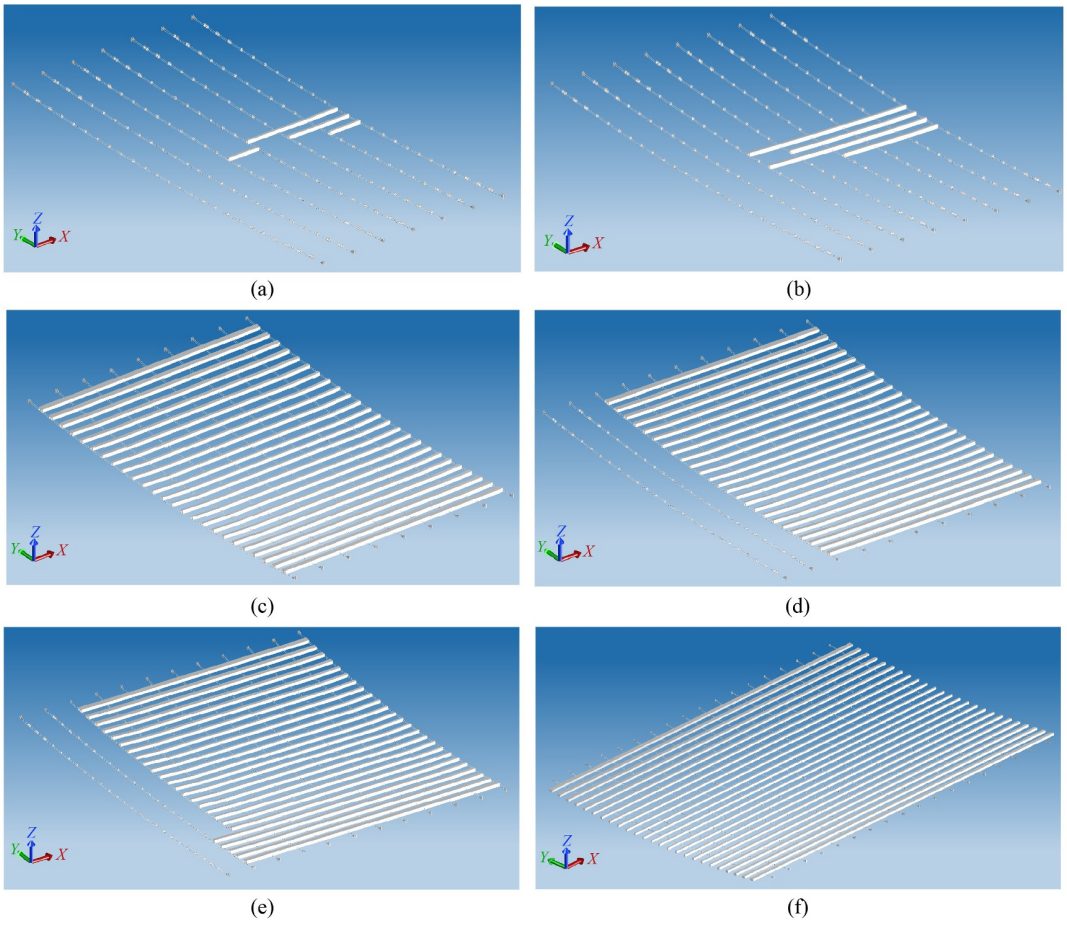


Fig. 14 Multi-stage IFC model: (a) IFC model on November 2, 2022; (b) IFC model on November 4, 2022; (c) IFC model on November 30, 2022; (d) IFC model on December 2, 2022; (e) IFC model on December 5, 2022; (f) IFC model at as-built stage.

1) Two cables from a set are simplified into a cable, with double cross-sectional areas of one single cable.

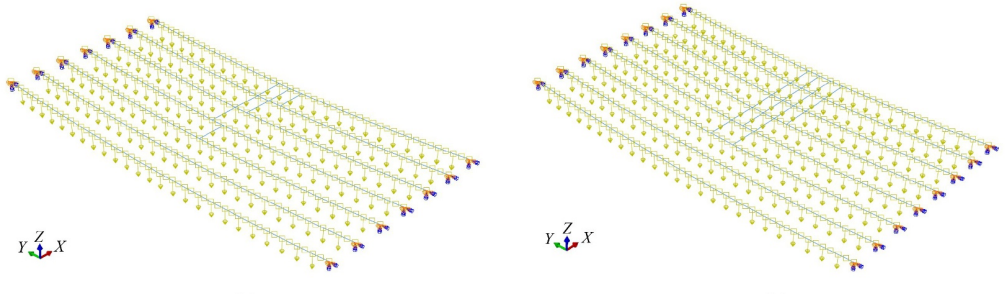
2) The loads of clamps are simplified as concentrate loads.

3) The displacements of anchor ends are zero during the construction.

4) The self-weight of curtain walls and other ancillary components are uniformly represented by increasing the density of the circular pipes which is illustrated in Table A7 at Appendix A in Electronic Supplementary Material.

Since the design spacing between cables is 8.4 m, the connecting members should be modeled as circular pipes with a length of 8.4 m. The final FEM at each moment of every stage scan is shown in Fig. 15.

This paper verified the difference between two parallel cables and a single cable simplified by doubling the cross-sectional area. In the model, the self-weight of the cable clamp was evenly distributed to the two cables. Establish a coupling constraint between the two cables at the cable clamp position to keep their displacement changes consistent. Meanwhile, the original cross-sectional area of a single cable was adopted, and the corresponding temperature load was applied to induce the internal force of the cables. From the calculation results, it can be observed that the cable force (326.3 kN) of each individual cable is approximately equal to half of the



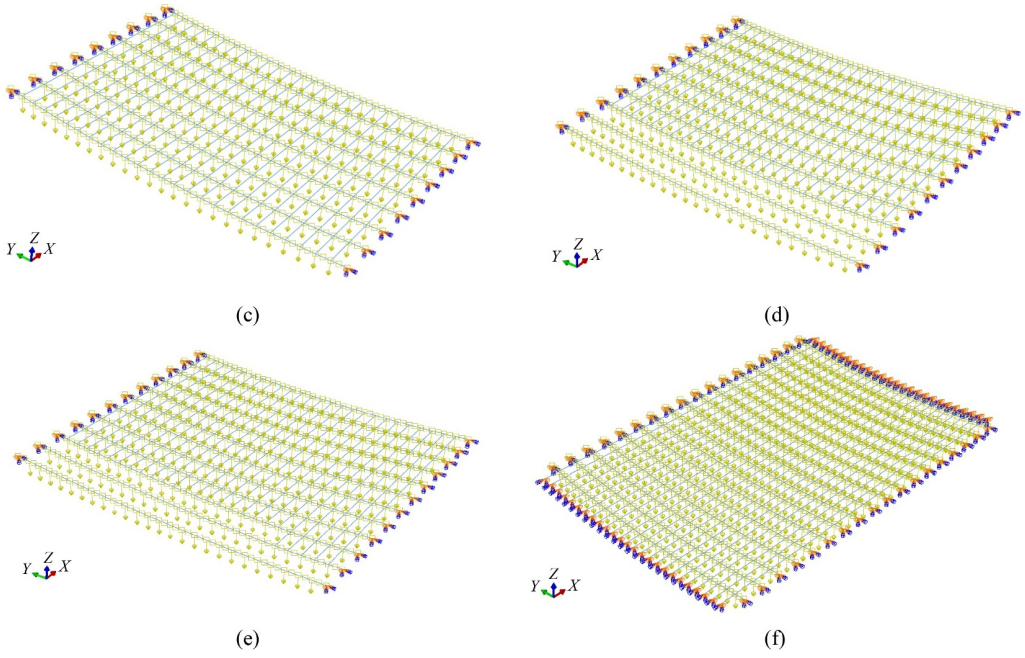


Fig. 15 Multi-stage FEM modeling: (a) FEM on November 2, 2022; (b) FEM on November 4, 2022; (c) FEM on November 30, 2022; (d) FEM on December 2, 2022; (e) FEM on December 5, 2022; (f) FEM at as-built stage.

cable force of the simplified single cable (652.7 kN) in Fig. 16. Additionally, the variation in cable shape is within 2 mm (max 1.557 mm), which proves that the model is also in a state of equilibrium and verifies the rationality of the simplification.

The paper also compared the results between distributed and concentrated clamp loads. The distributed load model is shown in Fig. 17. The cable force is 652.7 kN of distributed loads which is equal to the concentrated loads. Maximum deformation of distributed loads is 1.64mm which is basically consistent with concentrated loads (1.40 mm).

This paper conducts a sensitivity analysis on the influence of anchor displacement changes on the mid-span sag and cable force, with the results in Table 2. It can be seen that when the anchor end displacement is within 10 mm, its impact on both cable force and cable shape is within 5%.

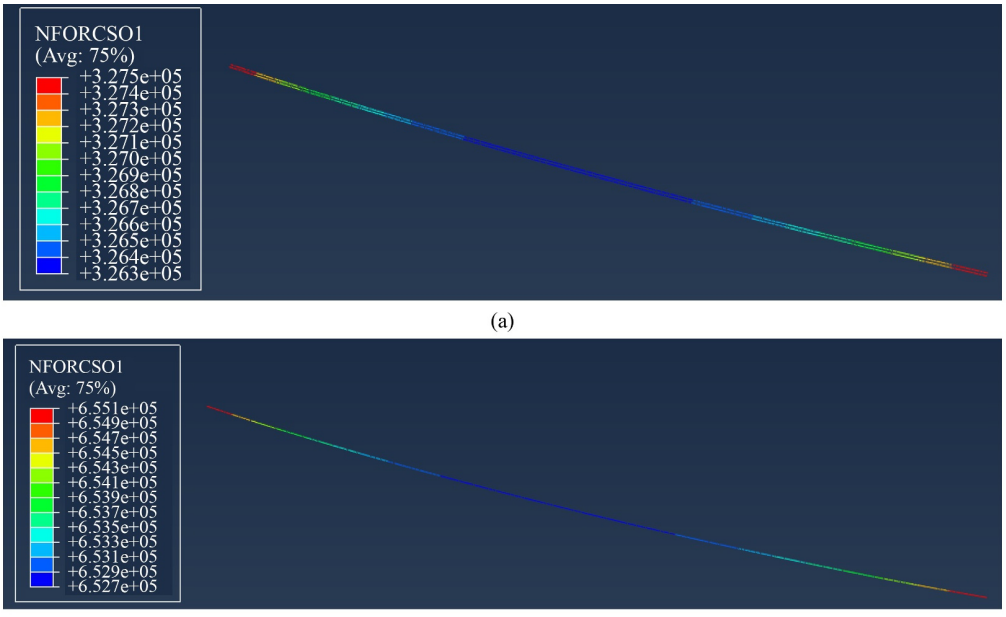
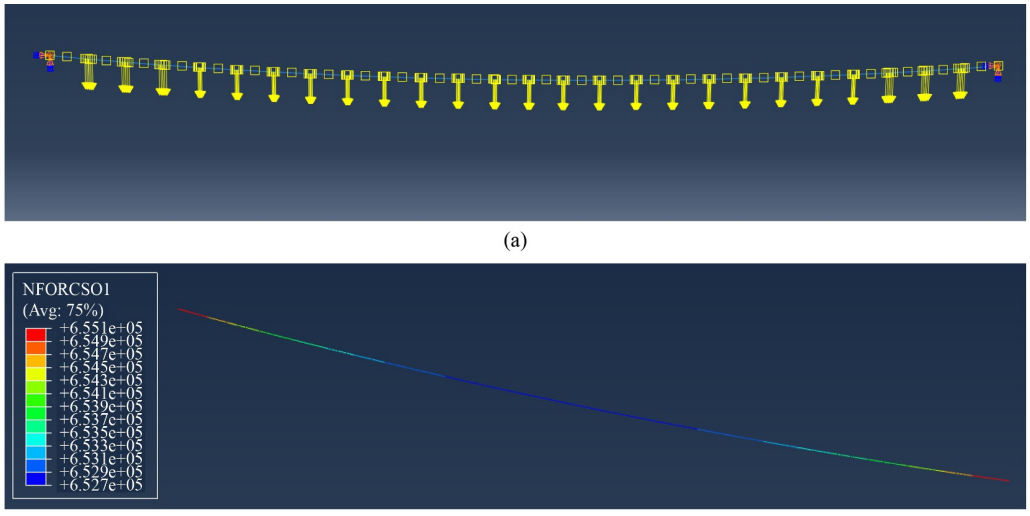


Fig. 16 Comparison between parallel cables and single cable with double cross-sectional areas: (a) parallel cables; (b) single cable with double cross-sectional areas.



(a)

(b)

Fig. 17 Result comparison between distributed loads and concentrated loads for cable clamps: (a) distributed loads for cable clamps; (b) results.

Table 2 Sensitivity analysis on the influence of anchor displacement changes

Displacement (mm)	Sag deformation (mm)	Sag (m)	Cable force (kN)	Force change (kN)
1	−7.5	2.2031	650.7	−2
5	−33.9	2.1767	643.0	−7.7
10	−66.9	2.1437	633.6	−19.1

4.2 Industry foundation classes and finite element model accuracy and efficiency

To test the accuracy of the established IFC model, OBJ geometric files for multiple stages are extracted from the IFC. The OBJ files and point clouds are then imported into CloudCompare [43] to calculate the average distance between the point clouds and the OBJ geometric model. The modeling accuracy of all models is presented in Table 3. Root Mean Square Error (*RMSE*), standard deviation, max/min errors and number of vertices/elements of point-to-model distances are supplemented in Table A8 at Appendix A in Electronic Supplementary Material. The Point-to-model distances histogram of 1102 stage is supplemented in Fig. 18. The histogram patterns at other stages are basically consistent. From the histogram, the proportion of point clouds with a distance greater than 0.1 m is basically less than 0.1%, so the maximum distance in the table is usually the distance between noise points and the BIM model.

As shown in Table 3, the modeling accuracy of the IFC is approximately 15–20 mm. The modeling time increases with the number of components, with the fully constructed structure requiring the longest modeling time, approximately 27.59 s. The main reasons for modeling errors are as follows.

1) In this study, the segmental catenary theory is used during the extraction of cable shapes, assuming that the cables do not undergo lateral deformation. In reality, the structure may still exhibit minor horizontal deformations.

2) The impact of noise or edge drift during the scanning process.

3) The extracted cable shape represents an average of two cables from a set of cables, and the actual shapes of the two cables may not be perfectly parallel.

4) The impact of precision inaccuracy in extracting key coordinates.

The modeling and computation times for the FEM are shown in Table 3. As construction components increase, the modeling time grows from 1.12 to 11.93 s, while the computation time increases from 36 to 98 s. The calculation result changes in cable shape and cable forces

Table 3 Model accuracy and efficiency

Date	IFC model accuracy (mm)	IFC modeling time (s)	FEM modeling time (s)	FEM calculating time (s)	Total time (s)
2022.11.02	16.7	3.93	1.12	36	41.05
2022.11.04	18.5	5.01	1.03	30	36.04
2022.11.30	19.9	8.03	3.01	54	65.04
2022.12.02	20.4	10.28	3.23	56	69.51
2022.12.05	19.7	11.94	3.95	60	75.89
Final stage	15.3	27.59	11.93	98	137.52

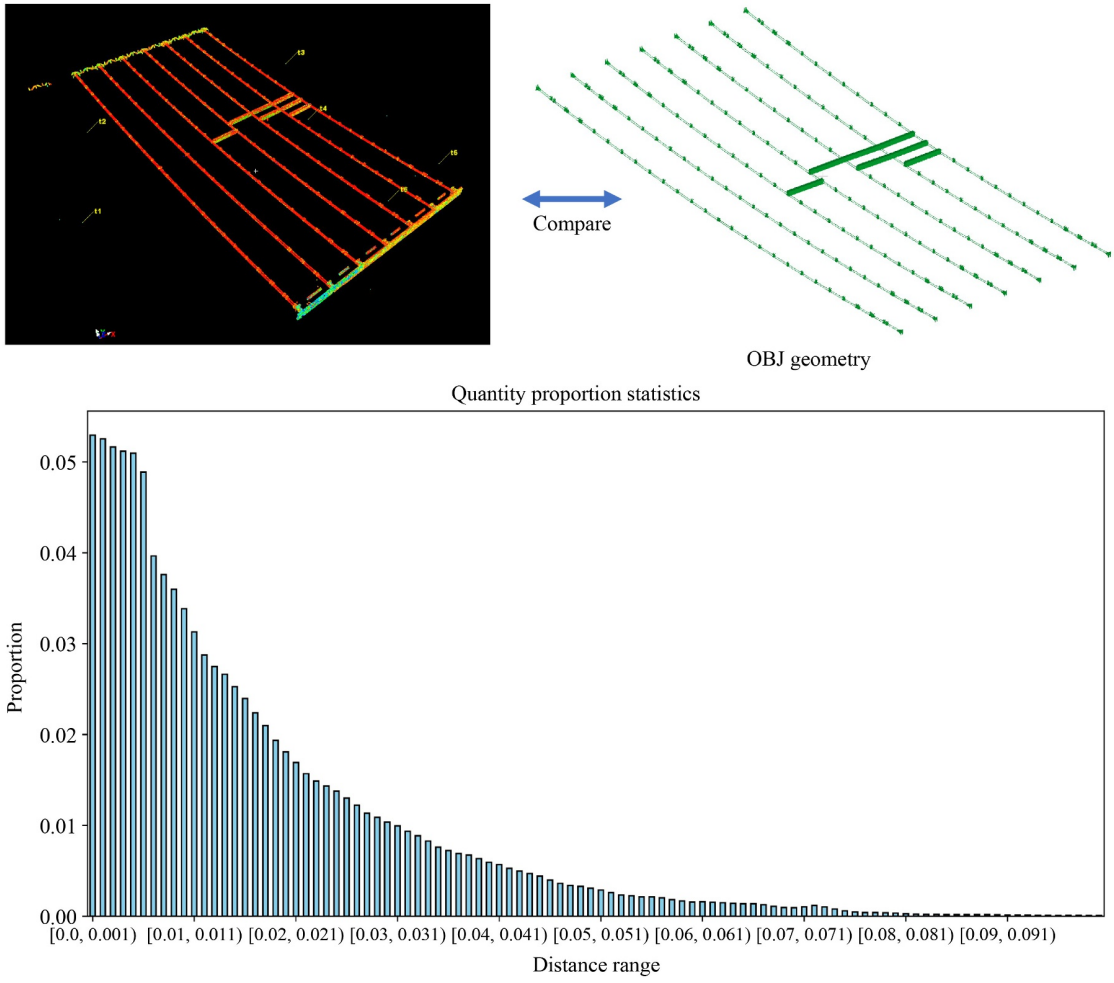


Fig. 18 Quantity proportion statistics of distances from points to BIM.

are presented in Tables A1–A6 at Appendix A in Electronic Supplementary Material. Based on the FEM calculated for the cable shape, the maximum calculated deformation of the FEM at each stage is within 4 mm. The maximum deformation of the cable relative to the sag is less than 0.2%, indicating that the cable deformation results are minimal. The cable force variations are within 0.5%. This demonstrates that the FEM is generally in equilibrium, suitable for subsequently predicting cable shape and forces based on temperature and construction progress.

4.3 Predicting cable shape and cable force considering temperature data and construction schedule

During the construction phase, the primary loads affecting the cable structure are temperature loads and the self-weight of the installed connecting members. Therefore, after establishing a FEM at one scan, finite element calculations can be carried out based on on-site temperature data and the recorded construction progress schedule.

In this section, the FEM established on November 4th is used as the initial calculation reference. By varying the temperature load and applying the self-weight of the connecting members, the finite element results are compared with the cable shape measurement results and cable forces from November 30th. The FEM is modeled based on the cable shape measurements from November 4th. The temperature loads are based on on-site temperature data: the average temperature during the scanning period from 3:15 PM to 4:16 PM on November 4th was about 17 °C, while the average temperature during the scanning period from 5:20 PM to 9:00 PM on November 30th was about −0.15 °C. Therefore, after applying the cooling method to adjust the cable forces, a temperature load of −17.15 °C is added to reflect the temperature difference. Additionally, the connecting members are assembled according to the installation status of November 30th. Since the 4th cable was not installed on November 4th, the connecting members between the 4th and 5th cables could not be assembled. Therefore, half of the self-weight of the connecting members at different positions is applied as a concentrated load at the cable clamp position of the 5th cable.

The deviations between the finite element calculations and actual measurements of the cable shapes in November 30th are shown in Table 4. It can be observed from the table that the prediction deviations by the finite element method are generally within 3 cm, with a percentage deviation of less than 1%. The deviation for the 10th cable is relatively large, around 3.8 cm. This discrepancy may be due to deformations at the anchorage ends under temperature load effects, or inaccuracies in the self-weight load of the connecting members. The deviation in cable forces is also within 2%, which generally meets engineering requirements. When moving to the next scanning phase, the FEM can be re-established base on new scan geometry to correct the deviations from the previous calculations.

4.4 4-Dimension industry foundation classes visualization based on the calculation results from finite element model

Based on the above method, continuous updating of the IFC model between scanning phases can be achieved. The geometric data are from the cable shapes after deformation according to the FEM calculations. For example, starting from November 4th, Fig. 10 shows the positions of the connecting members installed daily between cable 6–cable 12 from November 5th to November 10th. Based on this record, the finite element and IFC models can be updated on a daily basis.

Based on the daily construction records, the finite element modeling process can be adjusted by modifying the connecting member quantity matrix. Finite element modeling and calculations should be conducted in Subsection 2.2. Using Abaqus for secondary development allows for the rapid extraction of deformed cable shapes from the ODB files. Afterwards, the IFC model can be constructed as described in Subsection 2.1 based on these deformed cable shapes, resulting in an evolution of the construction process on a daily basis. The IFC models from November 5th to November 10th are shown in Fig. 19. Since no connecting members were installed on November 8th and there was minimal shape variation during scanning on the day, the IFC model for November 8th is not displayed.

5 Conclusions

This paper presents a method for creating a dynamic gDT for a cable-net structure in the construction process. Considering the issue of update synchronization between BIM and FEM, the parametric modeling of the IFC model and FEM is achieved based on the common geometric information extracted from multi-stage scans. Considering the issue of information loss between two scanning stages, the FEM of the previous stage is used as a starting point, and the calculation results based on construction progress and temperature data are integrated to supplement the deficient structural geometric information. The creation of a continuous dynamic IFC model is achieved based on construction progress and deformed cable shapes that are derived from finite element calculations. The proposed method can serve for the construction simulation, cable force prediction and visualization of construction process of cable structures.

Regarding the accuracy of IFC modeling, the average distances between the parametric multi-stage BIMs and point clouds are from 15 to 20 mm. The calculated deformation of the FEM at each stage is within 4 mm, demonstrating that the established model is in a state of equilibrium. The total duration for modeling the IFC model, FEM, and computation for each stage is within 140 s. The prediction deviations of cable shapes and forces between the FEM results from the previous stage and measurement results of the next stage are within 1.3% and 1.5%, respectively.

The proposed workflow can be applied to larger or more complex cable-net structures. The modeling process is only based on node coordinates and connection relationship determination. Once the geometric relationship between the cables and other components is determined, the geometric models of the cables can be realized by approximating curved elements with straight segments, and other components can also be parametrically modeled based on this relationship. The proposed

Table 4 Deviation between FEM calculation results and measured values

No.	11.4 mid-span sag (m)	Deformation (m)	FEM calculation sag (m)	11.30 measurement sag (m)	Cable shape deviation	FEM calculation force (kN)	11.30 cable force (kN)	Cable force deviation
5	2.1871	−0.9879	3.1750	3.1581	−0.0169	1901.133	1917.506	0.854%
6	2.1956	−0.9735	3.1691	3.1510	−0.0181	1889.665	1911.568	1.146%
7	2.3037	−0.8310	3.1347	3.1426	0.0079	1923.124	1921.929	−0.062%
8	2.4780	−0.6385	3.1164	3.1272	0.0107	1951.155	1944.528	−0.341%
9	2.5801	−0.5149	3.0950	3.0860	−0.0089	1945.403	1947.385	0.102%
10	2.5671	−0.4816	3.0487	3.0866	0.0379	1969.858	1940.997	−1.487%
11	2.5857	−0.4758	3.0615	3.0572	−0.0043	1953.380	1966.745	0.680%
12	2.3922	−0.2305	2.6228	2.6109	−0.0118	1430.293	1420.214	−0.710%

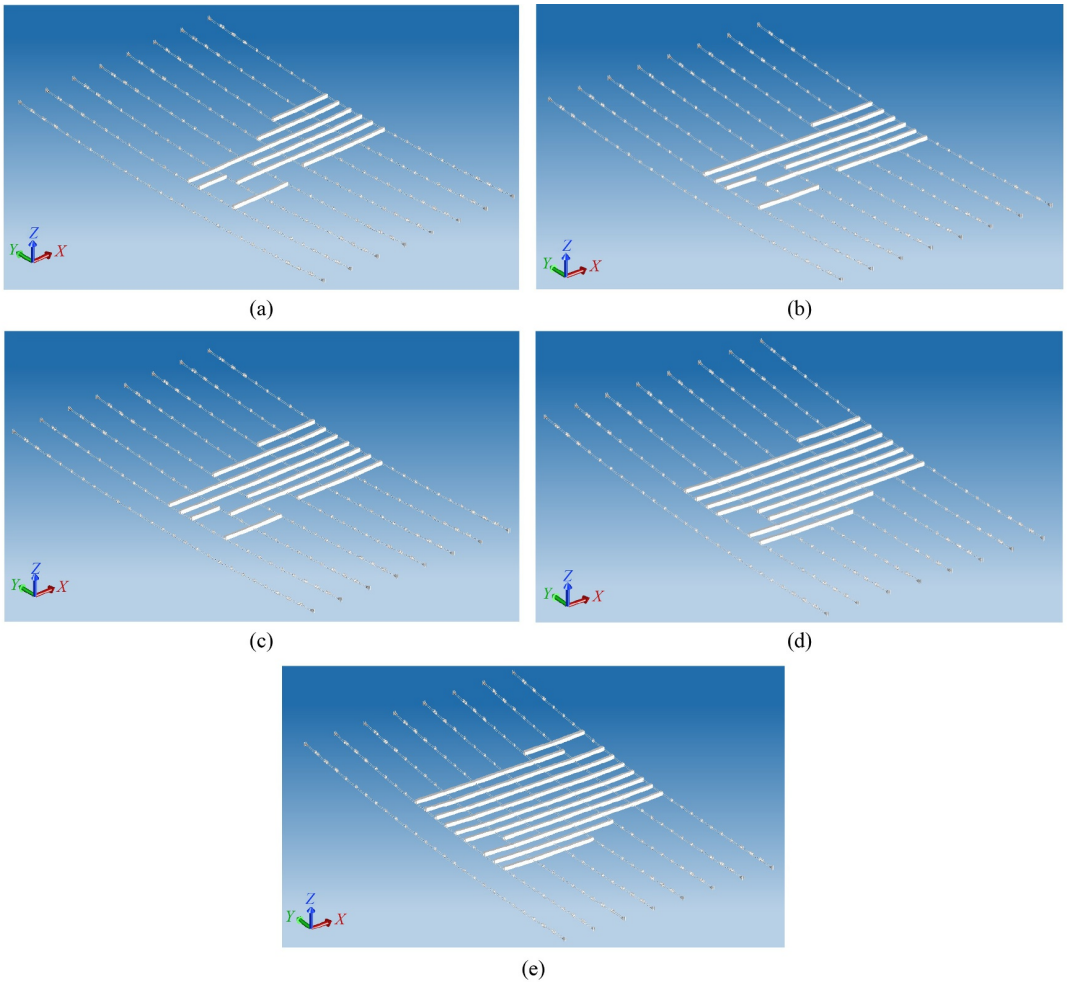


Fig. 19 Continuous updating of the IFC model during construction: (a) IFC on November 5, 2022; (b) IFC on November 6, 2022; (c) IFC on November 7, 2022; (d) IFC on November 9, 2022; (e) IFC on November 10, 2022.

FEM model utilizes beam elements, which not only significantly reduce computational complexity but also keep consistency with FEM models for larger-scale cable structures. The finite element calculation approach achieves the equilibrium state of the cable net structure by applying external loads and internal cable forces. It is also applicable to other types of cable net structures.

There are still some limitations in the work presented in this paper that can be improved in the future. For example, the current connecting members are only considered as a rectangular prism, and the complex internal components are not included. The IFC model includes only geometric data currently, while other information such as material data, load information, and construction information can be supplemented in IFC data. The FEM in this paper assumes that the displacement deformation at the cable anchorage is zero, and its boundary stiffness can be adjusted based on monitoring data or other information.

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Competing interests The authors declare that they have no competing interests.

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